

ANKIT KUMAR TIWARI

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SUMMARY

Dedicated Computer Science Engineering student at VIT Bhopal with a strong focus on Machine Learning and Deep Learning. Proficient in developing AI-driven solutions, including real-time computer vision systems and containerized ML pipelines. Active competitive programmer on Codeforces and LeetCode, passionate about solving complex algorithmic challenges and building scalable, data-centric applications.

TECHNICAL SKILLS

- **Data Science & ML:** Python, Pandas, Numpy, scikit-learn, Tensorflow, Pytorch, Feature Engineering, Model Evaluation, Clustering, Statistical Analysis, Regression
- **AI & NLP :** LangChain, RAG Pipelines, Transformers, LLM APIs(OpenAI/ Anthropic/Google), Text Classification, NLP preprocessing
- **Data Visualisation:** Matplotlib, Seaborn, Plotly, Leaflet.js, REST API Dashboards, Interactive Charts and Reports
- **Automation & Dev:** FastAPI, React.js, Next.js 14, Node.js, ReactNative, MySQL, Flask, Rest APIs, JSON, Git/Github

EDUCATION

VIT Bhopal University

Aug 2024-May 2028

B.Tech in Computer Science & Engineering — CGPA: 8.73 / 10.0

Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, DBMS, Operating Systems, Computer Networks, Statistics, Object-Oriented Programming

PROJECTS

Music Recommendation System | Python, NLP, scikit-learn, RAG Pipeline, Flask, REST API

- Performed end-to-end exploratory data analysis on diverse music metadata; engineered features for similarity scoring and content classification using NLP and ML algorithms, significantly improving retrieval precision and model relevance.
- Built, trained, and rigorously evaluated multiple predictive models for music retrieval; deployed the best-performing model as a Flask REST API achieving sub-100ms response latency with full pipeline documentation for team knowledge transfer.
- Designed reusable ML pipeline templates and automated preprocessing workflows ensuring reproducibility, consistency, and scalability across varied and large-scale dataset inputs

YouTube Video Chatbot | Python, LangChain, FAISS, RAG, HuggingFace Llama 3, Streamlit

- Built a RAG pipeline that extracts YouTube transcripts, generates semantic embeddings (BGE-large), and stores them in a FAISS vector store for similarity-based context retrieval.
- Integrated Meta Llama 3 via HuggingFace API with LangChain prompt templates and parallel runnables to deliver context-grounded conversational Q&A over video content.
- Deployed an interactive Streamlit web app with modular architecture and session-state management, enabling end-to-end video-to-answer functionality from a single URL input.

AI-Powered Career Coaching System | Python, HuggingFace, Gemini API, Docker, Next.js, RAG

- Fine-tuned a DistilBERT NLP model with PyTorch to categorize user text into 65 distinct career paths.
- Engineered a RAG-based backend using the Gemini API to provide personalized career advice and generate interview questions based on user input.
- Built the frontend chatbot interface using Next.js and containerized the full-stack application with Docker Compose.

EXTRACURRICULARS & PARTICIPATIONS

- **National Hackathon Finalist** — SIH & India Innovates
- **Core Member, Linpack Club, VIT Bhopal**
- **Interests:** Cricket and Technical Problem Solving.
- **Competitive Programming:** Codeforces Rating(1000+)